Possibility (Sample Space) Diagrams

Exam code: 0607

Download PDF

Written by: Mark Curtis
Reviewed by: Dan Finlay
Updated on 26 November 2025

Possibility Diagrams

What is a possibility diagram?

- In probability, the **sample space** means **all the possible outcomes**
- In simple situations it can be given as a **list**
  - For flipping a coin, the sample space is: Heads, Tails
    - the letters H, T can be used
  - For rolling a six-sided dice, the sample space is: 1, 2, 3, 4, 5, 6
- If there are **two** sets of outcomes, a **grid** can be used
  - These are called **possibility diagrams** (or sample space diagrams)
  - For example, roll two six-sided dice and add their scores
  - A list of all the possibilities would be very long
    - You might miss a possibility
    - It would be hard to spot any patterns in the sample space

2nd dice

1st dice	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

- Combining **more than two** sets of outcomes must be done by **listing** the possibilities
  - For example, flipping three coins

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

How do I use a possibility diagram to calculate probabilities?

- Probabilities can be found by **counting** the **number** of possibilities you **want**, then **dividing** by the **total** number of possibilities in the sample space
  - In the sample space 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, there are four prime numbers (2, 3, 5 and 7)
    - The probability of getting a prime number is $\frac{4}{10} = \frac{2}{5}$
  - Using the possibility diagram above for rolling two dice, the probability of getting an eight is $\frac{5}{36}$
    - There are 5 eights in the grid, out of the total 36 numbers
- **Be careful**, this **counting** method **only** works if all possibilities in the sample space are **equally likely**
  - For a **fair** six-sided dice: 1, 2, 3, 4, 5, 6 are all equally likely
  - For a fair (**unbiased**) coin: H, T are equally likely
  - Winning the lottery: Win, Lose are are not equally likely!
    - You cannot count possibilities here to say the probability of winning the lottery is $\frac{1}{2}$

Examiner Tips and Tricks

- Some harder questions may not say "by drawing a possibility diagram" so you may have to do it on your own.

Worked Example

Two fair six-sided dice are rolled.

(a) Find the probability that the sum of the numbers showing on the two dice is an odd number greater than 5, giving your answer as a fraction in simplest form.

Answer:

Draw a possibility diagram to show all the possible outcomes

	1	2	3	4	5	6
1st dice						
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Circle the possibilities that are odd numbers greater than 5 (5 is not included)

2nd dice

	1	2	3	4	5	6
1st dice						
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Count the number of possibilities that are circled (12) and divide them by the total number of possibilities in the diagram (36)

$$\frac{12}{36}$$

Cancel the fraction

$$\frac{12}{36} = \frac{12 \times 1}{12 \times 3} = \frac{1}{3}$$

$$\frac{1}{3}$$

(b) Given that the sum of the numbers showing on the two dice is an odd number greater than 5, find the probability that one of the dice shows the number 2. Give your answer as a fraction in simplest form.

Answer:

From part (a) you already know there are 12 ways to get an odd number greater than 5

$$\frac{2}{12}$$

Cancel the fraction

$$\frac{2}{12} = \frac{2 \times 1}{2 \times 6} = \frac{1}{6}$$

$$\frac{1}{6}$$

Teacher only

Test your students on this in just a few clicks

- Create exam-aligned tests
- Tailor format & difficulty
- Quizzes or homework in minutes

Try Test Builder

Test yourself

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources

Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard

Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.